

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Name	Kishor Sreenivasan	Register No.	3122235001068
Due Date	19.02.2026		

Experiment 5: Perceptron vs Multilayer Perceptron with Hyperparameter Tuning

1. Objective

To implement and compare the performance of:

1. Single-layer Perceptron Learning Algorithm (PLA) (One-vs-Rest)
2. Multilayer Perceptron (MLP) with hidden layers and non-linear activation functions

Identifying the final tuned hyperparameters such as learning rate, activation function, optimizer, batch size, and number of hidden layers.

2. Dataset

English Handwritten Characters Dataset

- Total samples: 3,410
- Number of classes: 62 (0–9, A–Z, a–z)
- Image type: Grayscale

3. Preprocessing Steps

- Loaded the dataset using Pandas
- extracted the images from the Img folder
- Performed resizing, flattening and normalized the images
- Used LabelEncoder on the Labels column.

4. Implementation Details

- Performed Train and test split data.

- Built the Perceptron Learning Algorithm model from scratch.
- Trained the model and getting accuracy score and confusion matrix.
- Implemented PLA One vs. rest Approach and calculated accuracy, Epochs and Training loss.
- Identified the reason for Poor accuracy in PLA.
- Developed the Multilayer Perceptron Model
- Implemented 7 models with varying HyperParameters, with results documented as below
- Find the best tuned Hyperparameters and Model and compare with PLA.
- Developed the ROC curves of the Models.

5. Visualizations

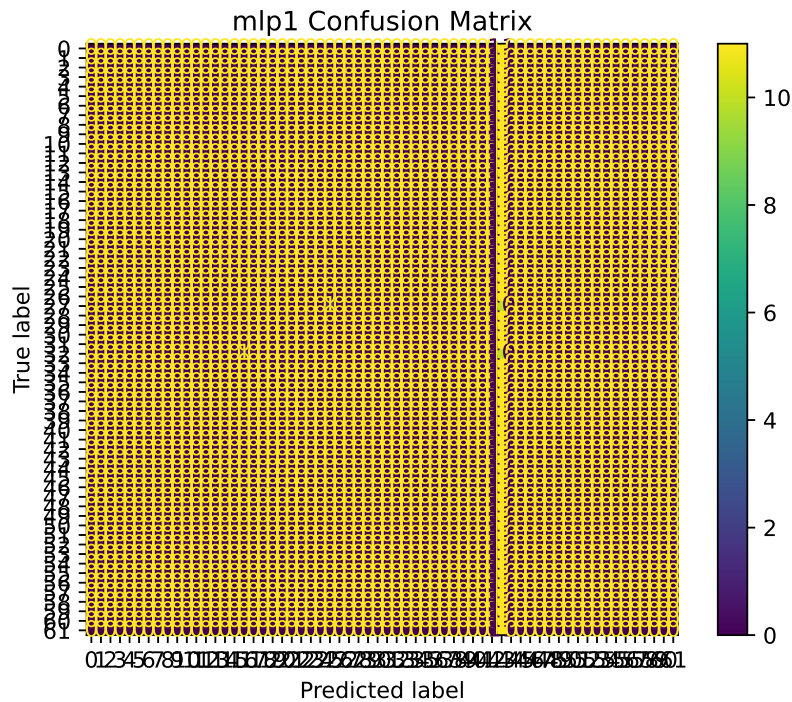


Figure 1: pla confusion matrix

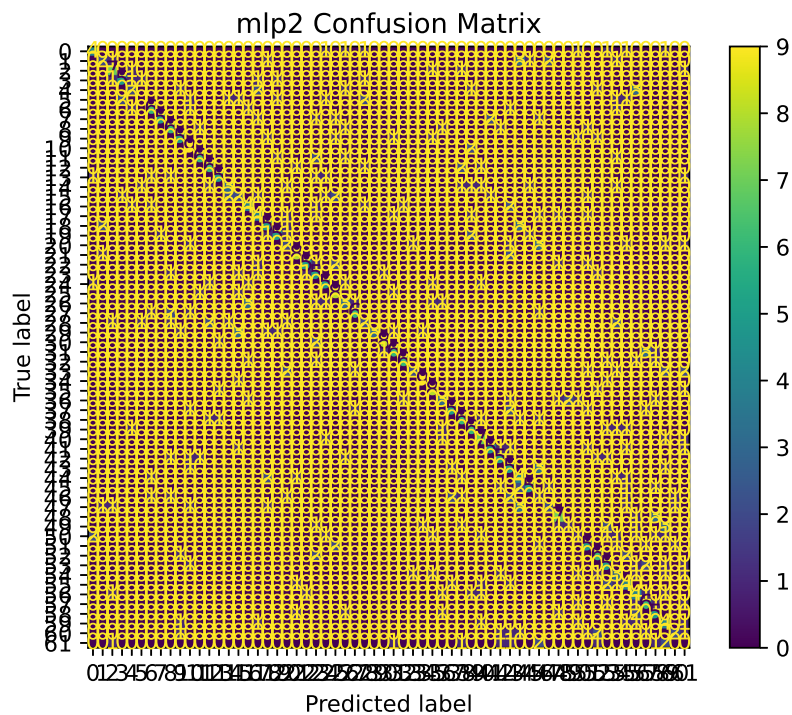


Figure 2: pla One vs Rest confusion matrix

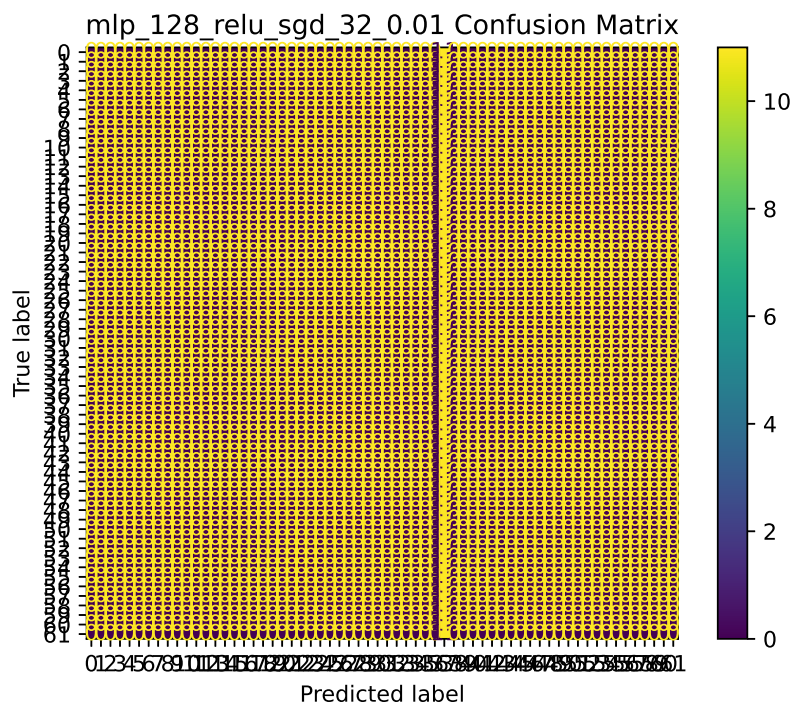


Figure 3: mlp1 confusion matrix

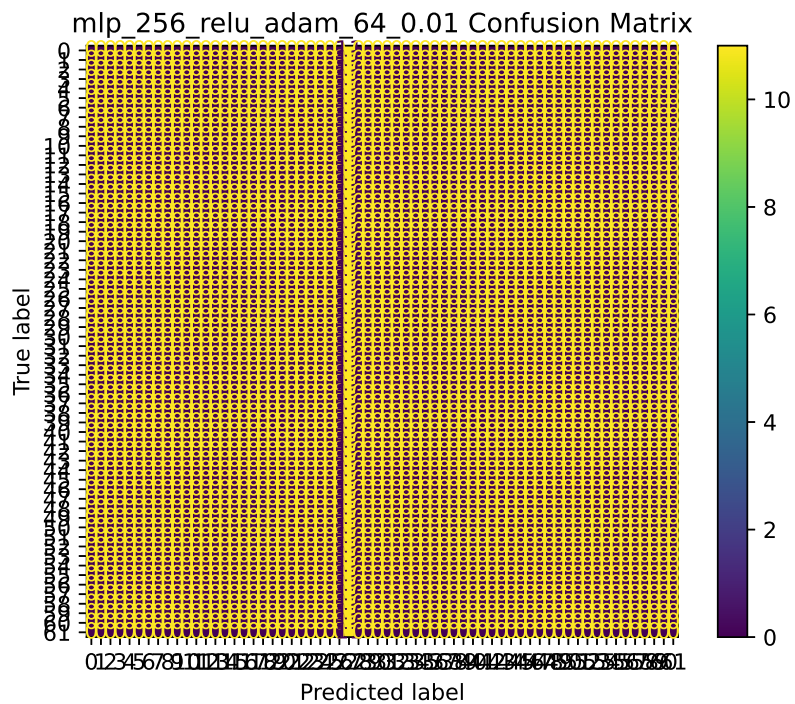


Figure 4: mlp2 confusion matrix

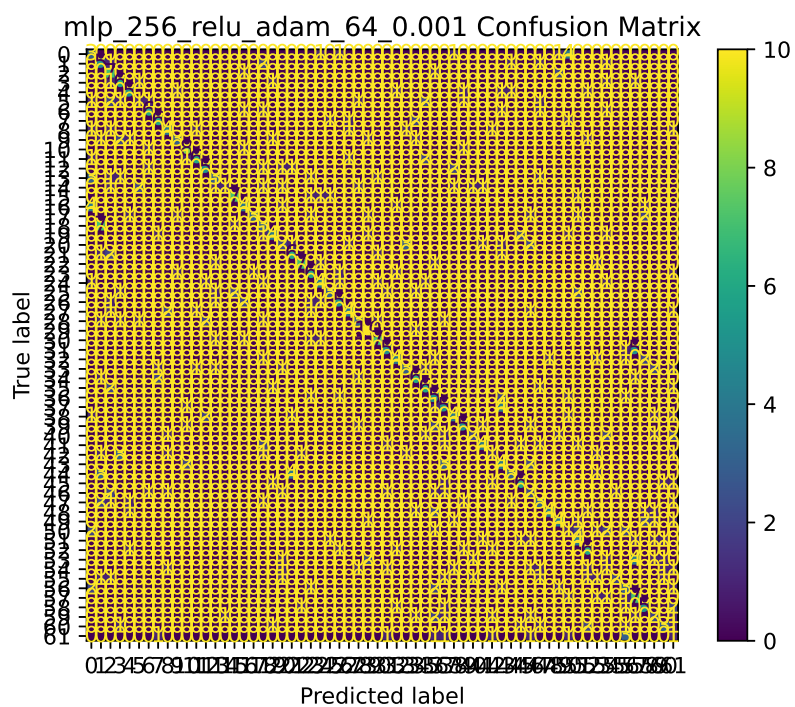


Figure 5: mlp3 confusion matrix

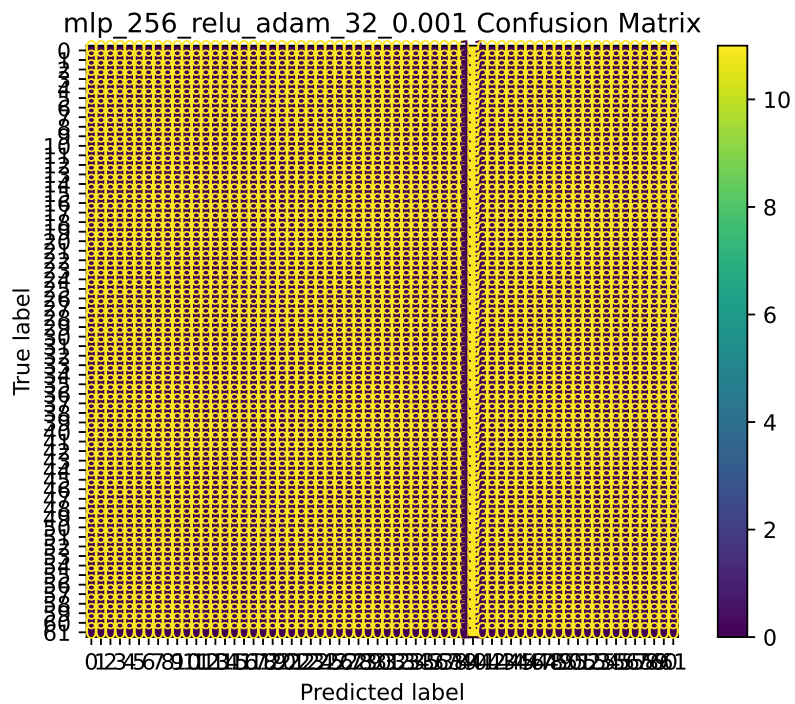


Figure 6: mlp4 confusion matrix

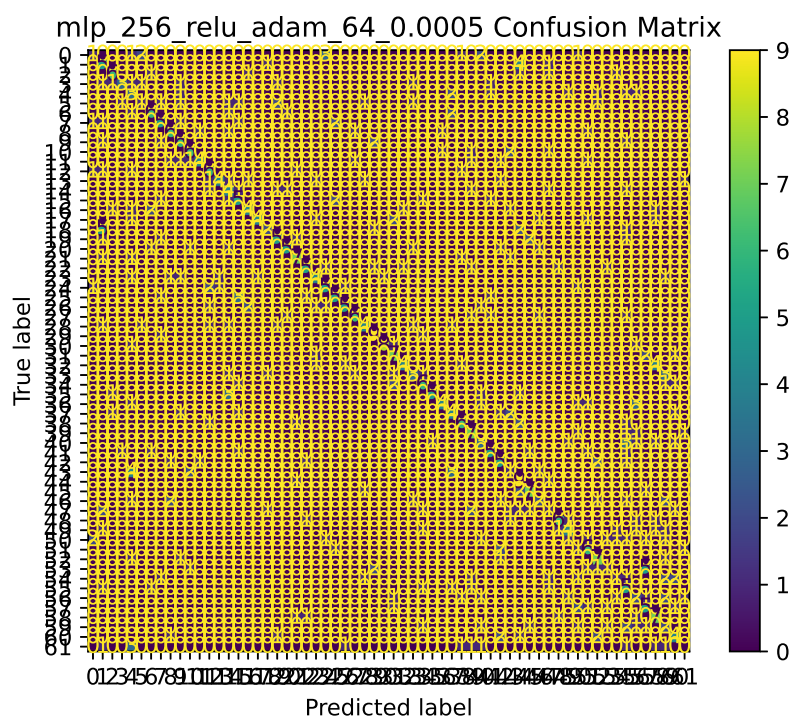


Figure 7: mlp5 confusion matrix

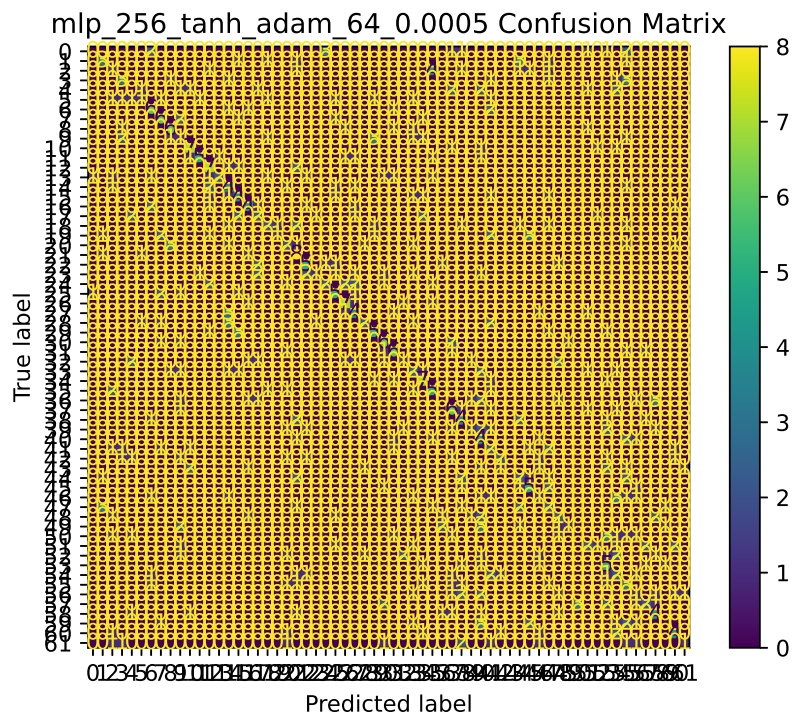


Figure 8: mlp6 confusion matrix

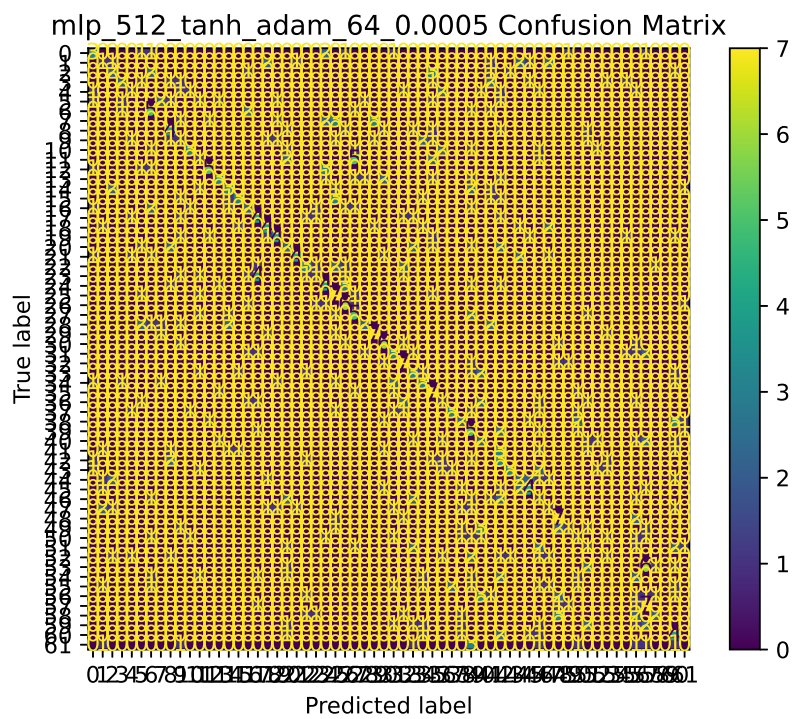


Figure 9: mlp7 confusion matrix

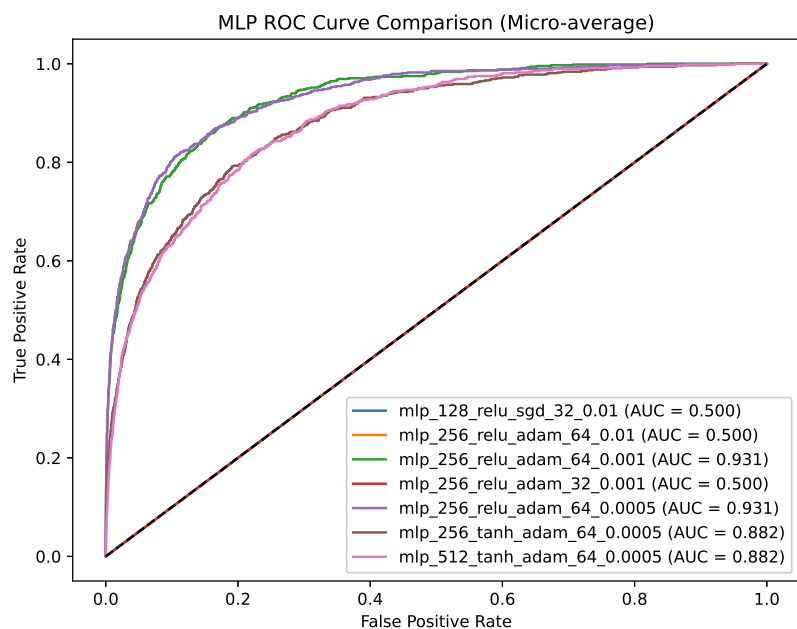


Figure 10: ROC Curves comparison

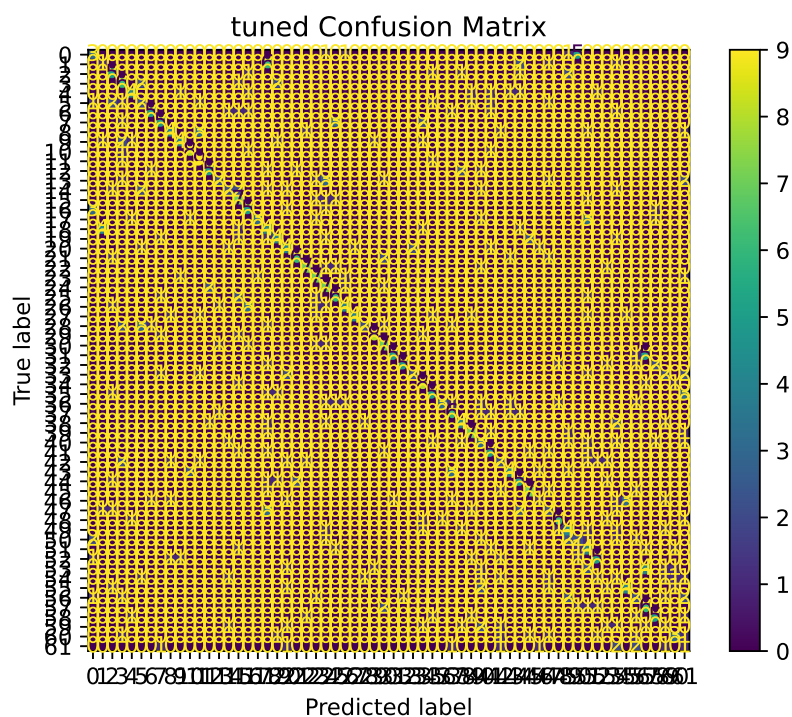


Figure 11: Tuned MLP confusion matrix

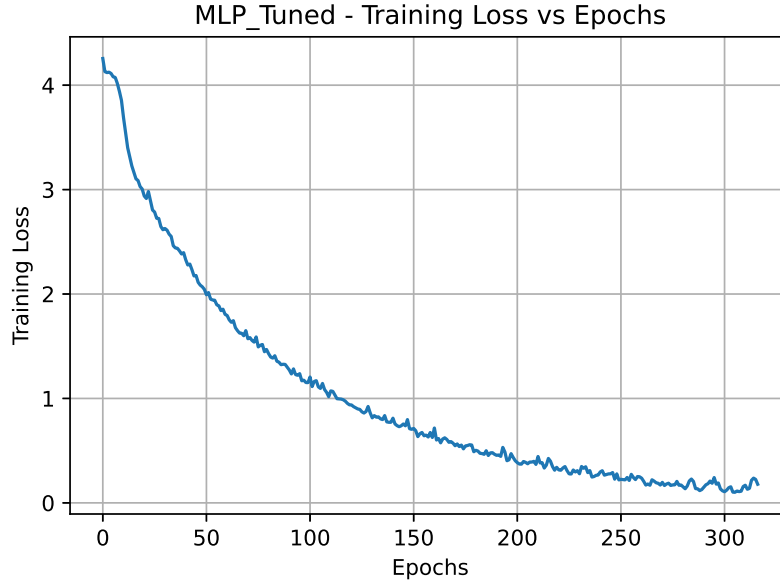


Figure 12: Training loss vs Epoch Curve

Overall Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-score
PLA (Scratch)	13.04%	0.1160	0.1304	0.0826
PLA (One-vs-Rest)	12.02%	0.2143	0.1202	0.0877
Tuned MLP	41.09%	0.4276	0.3929	0.3922

Hyperparameter Tuning Results for MLP

S.No	Hidden Layers	Activation	Optimizer	Learning Rate	Batch Size	Accuracy (%)
1	1 (128)	ReLU	SGD	0.01	32	1.6
2	2 (256-128)	ReLU	Adam	0.01	64	1.6
3	2 (256-128)	ReLU	Adam	0.001	64	41.00
4	2 (256-128)	ReLU	Adam	0.001	32	1.6
5	2 (256-128)	ReLU	Adam	0.0005	64	40.76
6	2 (256-128)	Tanh	Adam	0.0005	64	22.71
7	3 (512-256-128)	Tanh	Adam	0.0005	64	25.22

Note: There may be low accuracy due to maximum iterations set at 120, leading to incomplete convergence.

Best Tuned MLP Model

- Hidden Layers: 2 (256-128)

- Activation Function: ReLU
- Optimizer: Adam
- Batch Size: 64
- Learning Rate: 0.001
- Final Accuracy: 41.00%

Despite low accuracy compared to deeper models, this was due to incomplete convergence and limited iterations. It is the Highest and most stable model, as seen in the ROC curve comparison.

Training Convergence Comparison

Model	Epochs	Final Training Loss	Convergence Behaviour
PLA (OvR)	15.19 (avg)(the model never converges)	12.336	Non-convergent / Unstable
MLP (Tuned)	317	0.17824	Stable convergence

Strengths and Weaknesses of PLA vs MLP

Perceptron Learning Algorithm (PLA)

Strengths:

- Simple and easy to implement.
- Computationally inexpensive.
- Works well for linearly separable binary classification problems.
- Requires minimal hyperparameter tuning.

Weaknesses:

- Cannot learn non-linear decision boundaries.
- Performs poorly in multi-class classification tasks.
- Convergence is guaranteed only for linearly separable data.
- Does not provide probabilistic outputs.
- Sensitive to class imbalance in One-vs-Rest setup.

Multilayer Perceptron (MLP)

Strengths:

- Can learn complex non-linear decision boundaries.
- Handles multi-class classification effectively.
- Uses hidden layers to extract hierarchical feature representations.
- Provides probabilistic outputs (via softmax).
- Achieves significantly higher accuracy on handwritten character recognition.

Weaknesses:

- Computationally more expensive than PLA.
- Requires careful hyperparameter tuning.

- Risk of overfitting if network is too large.
- Training time increases with depth and number of neurons.

Effect of Hyperparameter Tuning on Convergence and Accuracy

The performance of the MLP model was highly sensitive to hyperparameter selection.

- **Learning Rate:** A learning rate of 0.01 caused unstable training and near-random accuracy (1.6%), indicating divergence or oscillation. Reducing the learning rate to 0.001 stabilized gradient updates and improved accuracy to approximately 41%. This demonstrates that learning rate has the most significant impact on convergence.
- **Optimizer:** Adam optimizer provided faster and more stable convergence compared to SGD. Adaptive learning rates in Adam helped in better optimization.
- **Batch Size:** Moderate batch sizes (64) provided stable gradients and balanced computational efficiency. Smaller batch sizes introduced noisy updates, while larger batch sizes slightly reduced generalization.
- **Activation Function:** ReLU performed better than Tanh for this dataset, as it avoids vanishing gradient problems and speeds up convergence.
- **Number of Hidden Layers:** Increasing depth improved representation learning up to a point, but deeper networks did not always guarantee higher accuracy due to limited dataset size and incomplete convergence.

Overall Observation: Hyperparameter tuning significantly improved the convergence behavior of the MLP model. Proper tuning transformed the model from near-random performance (1.6%) to stable convergence with substantially higher accuracy (41%). This highlights the importance of selecting appropriate learning rate, optimizer, and architecture for neural networks.

Learning Outcomes

- Learned the drawbacks of PLA in multiclass classification problems.
- Understood the differences between PLA and MLP models.
- Learned the effect of hyperparameters on model accuracy.
- Observed how learning rate, batch size, and activation functions play a major role.
- Built a PLA model from scratch.
- Calculated epochs and training loss on the dataset.

Conclusion

The Perceptron model underperforms in handwritten character recognition due to its linear nature and inability to learn non-linear decision boundaries.

The Multilayer Perceptron significantly improves performance through hidden layers, non-linear activation functions, and proper hyperparameter tuning.

Learning rate was found to be the most influential hyperparameter affecting convergence and final accuracy.